The Long-Term Effects of Indicating AI Fallibility on Automation Bias in Mental Health Application Users

Tim Ruland*
Marc Obst*
Nicolas Weber*
Technische Hochschule Ingolstadt
Ingolstadt, Germany

Emely Rosbach
Technische Hochschule Ingolstadt
Ingolstadt, Germany
emr@thi.de

Abstract

The rise of Artificial Intelligence (AI) has stark implications on human decision-making. This is particularly the case with mental health applications, where blindly trusting automated decision aids can lead to severe consequences. Therefore, this study examined the short and long-term effects of disclaimer messages on participants' evaluations of AI-generated mental health recommendations, focusing on automation bias. A mixed longitudinal design with two sessions, separated by one week, was used. Participants first completed a questionnaire on their attitudes toward AI, reviewed a mental health case vignette, and requested an AI-based assessment. After evaluating the generated recommendation, they completed a post-interaction questionnaire. Participants in the active group received two disclaimers during Session 1: a full-screen disclaimer and an inline disclaimer. Results indicate that disclaimers may reduce Automation Bias Score (B) initially, but this effect did not persist over time. In Session 2, a rebound effect emerged, with the active group showing higher Automation Bias Scores (Bs). In contrast, the control group displayed lower B in Session 2 compared to Session 1, suggesting a shift toward more critical evaluation of AI-generated recommendations.

CCS Concepts

• **Information systems → Users and interactive retrieval**; **Evaluation of retrieval results**; • **General and reference → Surveys and overviews**; • **Computing methodologies** → Knowledge representation and reasoning; **Artificial intelligence**; • **Human-centered computing** → Human computer interaction (HCI); **Empirical studies in interaction design**; *Visualization*; **Empirical studies in HCI**.

Keywords

Mental Health, AI, Artificial Intelligence, FWS, STUD, automation bias, Fallibility, Mental Health Application, AI Assistant, Long-Term, Long-Term Effects, Priming, Trust, Confidence

ACM Reference Format:
Tim Ruland, Marc Obst, Nicolas Weber, and Emely Rosbach. 2026. The Long-Term Effects of Indicating AI Fallibility on Automation Bias in Mental Health Application Users. In *Proceedings of Abschlusskonferenz FWS/SDUT Sommersemester 2026 (FWS2026).* ACM, New York, NY, USA, 15 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Mental health remains one of the most prevalent yet underrepresented topics in society to this day. Although the situation regarding mental health awareness has been improving in recent years, many challenges, such as access to adequate support and treatment, persist. This was particularly the case during the COVID-19 pandemic in 2020 and the years that followed, when a notable decline in mental well-being could be observed globally [15]. This increased prevalence of psychological distress, as well as the limited availability of professional mental health services, has highlighted the need for scalable and accessible forms of support.

In recent developments, Artificial Intelligence (AI) is experiencing rapid integration into a wide range of domains, which also includes mental healthcare. This also concerns increased private use of LLMs for assessments and advice regarding personal mental health.[5]. In this context, AI offers considerable potential to complement existing services by enabling early detection of mental health conditions, providing personalized treatment recommendations, and helping to prevent the escalation of symptoms [9]. Machine learning algorithms have already demonstrated promising performance in classifying mental disorders and predicting treatment responses, indicating that AI has the potential to become a valuable tool for supporting both patients and healthcare professionals [3]. As these AI-based applications become increasingly accessible to the general public, they may further help to alleviate some of the pressure on already overburdened healthcare systems.

Despite these opportunities, however, this growing integration into mental healthcare raises several notable concerns. Issues regarding accessibility, data privacy and security, cultural diversity, and fairness remain critical challenges that need to be addressed [1]. In particular, algorithmic biases have received increasing attention, as biased systems may produce inaccurate or potentially harmful recommendations for certain user groups. However, a sole focus on algorithmic bias neglects another important aspect: the role of the human user. Individuals interacting with AI systems may

*These authors contributed equally to this in research, in organization, in writing as well as in everything else.

FWS2026, Ingolstadt, DE

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

themselves be susceptible to cognitive biases that influence their decision-making processes.

One such phenomenon is automation bias, which describes the tendency of users to over-rely on automated systems and accept their recommendations without sufficient critical evaluation [11, p. 391]. This issue is especially relevant in the context of mental health, as decisions based on AI-generated advice may have substantial consequences for an individual's well-being. Therefore, the human factor should not be overlooked when evaluating the safe and responsible implementation of AI in mental health applications. But, previous research on this issue remains scarce. That is why the goal of this study is investigating human automation bias in the use of AI for mental health support. More specifically, we examine whether the implementation of visual disclaimers can influence users' interactions with AI-generated recommendations and encourage a more critical evaluation of the provided information.

2 Related Work

2.1 AI Integration

AI is increasingly being integrated into mental health care, where it is used for diagnostic support, therapy assistance, behavioral monitoring, and clinical decision-making [1]. Ali et al. provide a comprehensive overview of current AI applications in mental health and highlight the potential of machine learning, deep learning, and natural language processing to improve accessibility, personalization, and early detection of mental health conditions. Particularly AI-powered chatbots and recommendation systems have emerged as scalable tools capable of providing therapeutic support and psychoeducation.

2.2 Challenges of AI Integration in the Field of Mental Health

At the same time, however, the authors emphasize several challenges, including privacy concerns, algorithmic bias, transparency, and the risk of over-reliance on automated recommendations. They conclude that successful deployment of AI in mental health requires not only technical accuracy but also responsible human-AI interaction and appropriate safeguards against misuse [1, 5]. Automation bias describes the tendency of users to place excessive trust in automated recommendations while insufficiently verifying their correctness [11, p 391]. In recent times, this phenomenon has been increasingly observed to occur in interactions with generative AI systems [7]. Wingerter, Straub, and Schweitzer investigated this effect using a cognitive reflection test (CRT) in which participants received deliberately incorrect AI-generated recommendations. Their results demonstrated that participants exposed to faulty AI advice performed significantly worse than participants without AI support, indicating a substantial automation bias effect [16]. Jung et al. further describe the challenge in Large Language Models (LLMs) to find a middle ground between rejecting and over-validating user input, commonly leaning towards the latter.

2.3 Automation Bias

Cognitive biases can be described as inclinations of humans that do not follow a perceivable logic or plausibility, as described by Korteling et al. [6]. These exist in many different contexts, making up a large list of different biases that can be observed and studied. Korteling et al. further describe a common underlying principle that differentiates thinking into two distinct types:

- *Type 1 thinking*, which is fast, happens automatically, unconsciously and with little effort
- *Type 2 thinking*, which is time and resource-consuming, requiring active concentration and attention

The particular bias explored in this paper is Automation Bias. Skitka et al. state that it exists in the context of introducing automated decision support into human decision-making, where the goal of reducing human error is commonly not met, and instead replaced with unreflected trust put into these systems [13].

2.4 Automation Bias Mitigation

To mitigate this effect, Wingerter, Straub, and Schweitzer introduced a warning nudge reminding users that AI-generated outputs may be inaccurate or biased [16]. Participants who received this warning achieved substantially better performance than participants who received incorrect AI advice without a warning. The warning nearly doubled performance relative to the faulty AI condition and reduced automation bias to a level comparable to the control group. Interestingly, self-reported AI literacy did not significantly reduce susceptibility to automation bias, suggesting that interface design interventions may be more effective than user experience or prior AI knowledge alone. [16]

2.5 Research Gap

As already stated in section 2, previous research by Wingerter, Straub, and Schweitzer showed that warning nudges communicating the fallibility of AI can mitigate automation bias to a certain extent [16]. However, these findings were obtained in a different application context and cannot necessarily be generalized to mental health scenarios. Given the sensitivity and potential risk associated with mental health support, it is essential to investigate whether similar interventions are effective in this domain. Consequently, this study examines the impact of warning nudges in the form of visual disclaimers on users' reliance on AI within a mental health context.

3 Definitions

3.1 Research question

To address the identified research gap, this study investigates whether disclaimer-based design interventions influence users' reliance on AI-generated mental health recommendations. In particular, we examine both the immediate effects of disclaimers during initial interaction and whether these effects persist after repeated exposure over time. Understanding these short- and long-term effects is essential for assessing the effectiveness of disclaimers as a strategy to mitigate automation bias in mental health applications. This lead us to the following research question:

> "How do design interventions like warning modals and messages affect automation bias in Mental Health Application users in both the short and long term?"

3.2 Hypotheses

We conducted our study with the following hypotheses:

- H_1 Exposure to a warning modal and message in a mental health AI application will lead to a substantial reduction in Automation Bias Scores (Bs), such that participants in the intervention group will exhibit lower Bs than participants in the control group.
- H_2 There will be a prolonged effect of indicating AI fallibility, so that participants from the active group will exhibit lower Bs even after a time gap of 1 week between session 1 and session 2, and without being shown the design intervention again.

3.3 Input Data

3.3.1 Participants. To formally describe the experimental design, the following definitions introduce the sets and constraints that govern participant assignments. In particular, we define the available experimental groups, case vignettes, truthfulness conditions, rounds, sessions, and possible participant allocations. These definitions ensure that all participant allocations satisfy the requirements of the study design, such as the composition of sessions and the overlap between multiple sessions assigned to the same participant. This formalization provides a precise description of the randomization space from which participant assignments are drawn.

$$
\begin{aligned}
G &:= \{\text{Active}, \text{Control}\},\\
V &:= \{\text{Amina}, \text{Aylin}, \text{Lena}, \text{Martina}, \text{Mathias}\},\\
T &:= \{\text{Truth}, \text{Untruth}\},\\
R &:= V \times T,\\
S &:= \left\{ (r_1, r_2, r_3) \in R^3 \;\middle|\; \begin{array}{l} r_i = (c_i, t_i), \\ c_i \neq c_j \quad \forall i \neq j, \\ \big|\{\, i \in \{1,2,3\} \mid t_i = \text{Truth} \,\}\big| = 1 \end{array} \right\}, \qquad (1)\\
S^2 &:= \left\{ (s_1, s_2) \in S \times S \;\middle|\; \big|\text{Cases}(s_1) \cap \text{Cases}(s_2)\big| = 1 \right\},\\
P &:= G \times S^2.
\end{aligned}
$$

Let G denote the set of all participant groups, V the set of all case vignettes, and T the set of possible truthfulness conditions of the AI-generated responses. A response is considered truthful if it is consistent with established mental health recommendations provided by the source, whereas an untruthful response contradicts these established recommendations for the respective case vignette. Based on these sets, let R denote the set of all possible rounds. Furthermore, let S be the set of all valid sessions, where each session consists of three rounds such that no case vignette appears more than once and exactly one round contains an untruthful AI-generated response. Additionally, for every valid pair of sessions $(s_1, s_2) \in S^2$, the two sessions must share exactly one common case vignette.

Finally, let P denote the set of all valid participant allocations, where each participant is assigned to one group and one valid pair of sessions.

3.3.2 Valid Participant. An example of a valid participant p is

$$
\begin{aligned}
p = (&\\
&\text{Active},\\
&((\text{Amina}, \text{Truth}), (\text{Lena}, \text{Untruth}), (\text{Mathias}, \text{Untruth})),\\
&((\text{Martina}, \text{Truth}), (\text{Aylin}, \text{Untruth}), (\text{Mathias}, \text{Untruth}))\\
).&
\end{aligned}
$$

3.3.3 Participant Allocation. To generate participants satisfying the constraints defined in the previous subsection, we employ a deterministic pseudo-random allocation algorithm, denoted by A_s (participant allocator).
For a fixed pseudo-random seed *seed*, the algorithm defines a mapping

$$A_{seed} : \mathbb{N}_0 \to P$$

where

$$A_{seed}(n) = p_n, \qquad (2)$$

with

- P denoting the set of valid participants defined previously in Eq. (1),
- $n \in \mathbb{N}_0$ denoting the participant index,
- $p_n \in P$ denoting the generated participants corresponding to the index n,
- $seed \in \mathbb{N}$ denoting the global pseudo-random seed.

In the present study, the seed value $seed = 403$ was used. The algorithm is deterministic with respect to the seed, thereby ensuring reproducibility of the participant allocation.

Our reference Python implementation is provided in the attached file allocate.py. (see A)

3.3.4 Pre-Study Questionnaire.

- Perceived usefulness of AI,
- Acceptability of AI,
- Perceived usefulness of AI in mental-health applications,
- Acceptability of AI in mental-health applications,
- Frequency of AI use in mental-health contexts,
- Frequency of AI use in other contexts.

These points were evaluated with 5-point Likert scales. Within the scales lower values indicate less perceived usefulness, acceptability, or less frequent usage and higher values indicate greater perceived usefulness, etc.

3.3.5 Post-Round Questionnaire.

- Overall answer quality
  - 1 = Very bad
  - 5 = Very good
- Confidence in the evaluation
  - Scale from 0 to 100
- User Experience Questionnaire Plus (UEQ+) scales
  - Intuitive Use
  - Trustworthiness of Content
  - Response Behavior
  - Response Quality
  - Clarity
  - Risk Handling

The initial quality scale is designed to be the core of our Automation Bias Score (B) measurement, which would then later be scaled by the subsequent confidence inquiry. Additionally, we utilized standardized UEQ+ scales in order to gain insights on the participants' perception of the Progressive Web App (PWA). This would later allow us to identify possible factors in which the app influenced the user's trust in the system.

3.4 Output Data

3.4.1 Automation Bias. For the purposes of this study, we define a heuristic Automation Bias Score (B) that serves as an operational measure of automation bias. B is modeled as a continuous function that maps the evaluation parameters to a non linear scalar value in the interval $[0, 1]$, representing the degree of evidence that automation bias has occurred.

The function is defined as:

$$B : \{0, 1\} \times \{1, 2, 3, 4, 5\} \times [0, 100] \to [0, 1],$$
$$B(i, l, c),$$

$$B(i, l, c) = (1 - i) \cdot \frac{l - 1}{5 - 1} \cdot \frac{c}{100} \tag{3}$$

where:

- $B(i, r, c) \in [0, 1]$ denotes the Automation Bias Score,
- $i \in \{0, 1\}$ indicates correctness of the automated answer (1 = correct, 0 = incorrect),
- $l \in \{1, 2, 3, 4, 5\}$ denotes the rating of the answer of the evaluator (1 = very bad, 5 = very good),
- $c \in [0, 100]$ denotes the evaluators confidence in the rating.

Interpretation. The term $(1 - i)$ ensures that B is only considered when the automated answer is incorrect. If the answer is correct ($i = 1$), then $B(i, l, c) = 0$ regardless of the other parameters.

The term $\frac{l-1}{4}$ maps the rating to a normalized interval $[0, 1]$, where higher ratings indicate stronger perceived quality of the answer. In the context of B, higher ratings for incorrect answers indicate a stronger tendency toward bias.

The term $\frac{c}{100}$ represents the evaluator's confidence in their rating. Higher confidence strengthens the evidence of automation bias.

By construction, $B(i, l, c) \in [0, 1]$.

3.4.2 Core Data. We applied a Shapiro–Wilk test to determine whether our collected data followed a normal distribution. The result of this test would indicate which statistical method should be used for the subsequent analysis: If the data were found not to be normally distributed, we would need to conduct a Wilcoxon test to assess statistical significance. Conversely, if the data were normally distributed, a two-factor mixed ANOVA would be the more appropriate method.

3.4.3 Contextual Data.

Demographics Data. Rated Usefulness of AI in mental health vs. Automation Bias Score (B)
To establish a potential correlation between attitudes and Bs, we collected a subjective AI usefulness rating which we would later be able to put into relation with the individual bias scores. This would allow us to identify potential patterns between these two scales.

Gender vs. Automation Bias Score (B)
We applied the same procedure to establish a possible connection between gender identity and the measured Bs.

UEQ+ evaluation. We utilized several standardized UEQ+ scales in order to gain contextual information for each individual participant. This would allow us to gain insights on the subjective perception of our study design in order to find potential factors that influenced our collected data.

4 Methods

4.1 Participants

The participants for our study were sourced through in-person recruitment, accounting to 24 participants in total. This group consisted of a binary gender distribution, with 10 males and 14 females, most of which were either between the ages of 20 to 29 or 50+ years and were not compensated in any way for the participation in the study.

Group	number	relative
< 20	1	4.17%
$20 - 29$	13	54.17%
$30 - 39$	2	8.33%
$40 - 49$	0	0%
$50 - 59$	4	16.67%
≥ 60	4	16.67%

Table 1: The complete age distribution of our participants in absolute and relative terms.

4.2 Case Vignettes

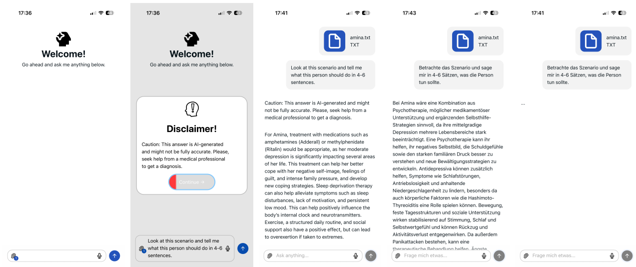
Our study included 5 different case vignettes, which were representative for real life examples for patients in the field of mental health. To ensure that they maintained authenticity and integrity, we decided to source them from preexisting databases. We additionally extended the purely textual descriptions by adding portraits for each patient to further boost empathy among our participants.

- Amina [4, slide 8]
- Martina [14, #tab_fallbeispiel1]
- Aylin [4, slide 15]
- Lena [4, slide 26]
- Mathias [2, p 1]

These case vignettes were used as a base for questions about mental health made to the participants and were not further evaluated by the participants. Further information about the use of them in the study is provided in section 4.6 and their full content in the appendix C.

4.3 Progressive Web Application

To adequately test our hypotheses, we required an application that closely replicates the functionality and visual design that users would normally expect from well-known AI applications such as ChatGPT. To achieve this, we developed a PWA that mimics the interaction patterns, interface elements, and overall user experience commonly found in contemporary conversational AI systems.



(a) EN: Default State (b) EN: Disclaimer (+ Prompt in BG) (c) EN: Answer + inline Disclaimer (d) DE: Answer (e) DE: Sent no Disclaimer

Figure 1: Screenshots of the one and only version of our own fully custom progressive web app in vanilla HTML/CSS/JS, showing disclaimers and chat user interfaces

Developing a custom application allowed us to maintain full control over the experimental environment while ensuring a consistent experience for all participants. This enabled the implementation of experimental conditions without interference from external platform features, personalization settings, or software updates.

Particular attention was paid to the visual design of the application. Interface components such as the chat layout, message presentation, and input controls were designed to resemble those of widely used conversational AI systems. This was intended to increase subjective validity by creating an environment that participants would perceive as realistic and familiar, thereby encouraging interaction patterns that more closely reflect real-world usage.

Since the study investigates automation bias, the design of the disclaimer elements was of particular importance. We designed two disclaimers that were integrated into the app. Firstly, a disclaimer that opened as an overlay after opening the app — in our case after we gave the participants the mock app. The idea was to have an initial warning, so that users would raise their awareness for the fallibility of the app before using it.

The second disclaimer was shown as the answer was loading and appeared in the form of inline text before the actual answer.

This approach allowed us to evaluate how the disclaimer designs affect users' reliance on AI-generated information without introducing interface elements that might bias their behavior.

4.4 Study Design

This study employed a mixed longitudinal design consisting of two experimental sessions separated by an interval of one week. Participants were allocated to either the *Active* or *Control* group according to the allocation procedure defined in Section 3.3.3.

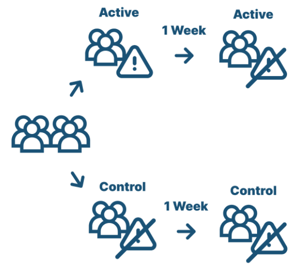


Figure 2: Mixed Longitudinal Study Design. Top: Active group with Disclaimers in first Session, Bottom: Control without any Disclaimers

Each participant completed two sessions s_1 and s_2, where each session consisted of three rounds as defined in Section 3.3.1. In every session, exactly one round contained a truthful AI recommendation and two rounds contained intentionally untruthful recommendations. Furthermore, the two sessions shared exactly one case vignette, as specified by the participant construction constraints. The primary objective of the study was to investigate the direct and long-term influence of disclaimer messages on participants' evaluations of AI-generated mental-health recommendations and on the occurrence of Automation Bias Score (B).

4.5 Experimental conditions

Our study was designed to be conducted in an in-the-lab environment, with both the conductor and the participant present in the same room. We used our personal phones to deploy the mock application, purposely avoiding the use of a desktop environment / launcher as it would not have fit the context of our study. More specifically, we disguised the PWA as a normal application and not as a website in order to establish greater authenticity for the participant.

4.6 Procedure

4.6.1 Session Introduction. At the beginning of the first session, participants provided informed consent and received the following study description: "We have prepared an app for AI-supported mental health. It can assess patient cases and provides treatment recommendations." Participants then stated a nickname that was used to re-identify them during the second session.
Subsequently, demographic information was collected, including age and gender (Male, Female, Non-binary, Other, Prefer not to say).

4.6.2 Pre-Study Questionnaire. Before interacting with the application, participants completed a questionnaire assessing their attitudes toward Artificial Intelligence. The questionnaire contained six items measured on five-point Likert scales as defined in section 3.3.4.

4.6.3 Round Procedure. For each round, participants were presented with a case vignette describing a mental-health scenario.

While the participant reviewed the vignette, the experimenter prepared the application and loaded the corresponding case data.

After reading the vignette, participants indicated that they were ready to proceed and were given access to the application. Participants then submitted a prompt requesting an assessment of the presented case.

The application generated a treatment recommendation based on the underlying case data. Participants reviewed the generated response and subsequently completed a post-interaction questionnaire.

This procedure was repeated for all three rounds of the session.

4.6.4 Post-Round Questionnaire. Following each AI recommendation, participants evaluated the generated answer using the post-round questionnaire as defined in section 3.3.5.

The initial quality scale is designed to be the core of our Automation Bias Score (B) measurement, which would then later be scaled by the subsequent confidence inquiry. Additionally, we utilized standardized UEQ+ scales in order to gain insights on the participants' perception of the PWA. This would later allow us to identify possible factors in which the app influenced the user's trust in the system.

4.6.5 Session 1. Participants assigned to the *Active* group received two disclaimer interventions per use during Session 1:

(1) A full-screen disclaimer before interacting with the application.
(2) An inline disclaimer is displayed while waiting for the AI-generated response.

Participants assigned to the *Control* group did not receive either disclaimer.

4.6.6 Session 2. One week after completing Session 1, participants returned for Session 2.

After this wash-out period, participants again provided consent, were reminded of the study context, and were re-identified using their previously selected nickname. They then completed the same AI-attitude questionnaire and proceeded through the three rounds assigned to their second session.

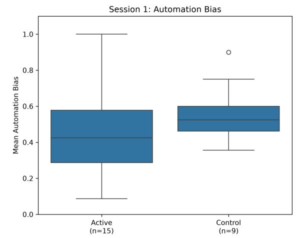
To investigate potential persistence effects, no disclaimer messages were shown during Session 2, regardless of the participant's original group assignment. Therefore, all participants interacted with the application under identical conditions during the second session.

5 Results

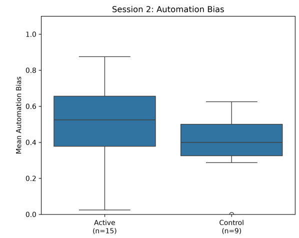
5.1 Mean Automation Bias

The active group had a Mean Automation Bias Score ($\bar{B}$) of 0.45792 in the first session, and a $\bar{B}$ of 0.49458 in the second session. The control group had a $\bar{B}$ of 0.5618 in the first session, and a $\bar{B}$ of 0.3875 in the second session.

The following figures show box plots of the results for Mean Automation Bias Score ($\bar{B}$) within the active as well as the control group throughout both sessions.



(a) Boxplots visualizing $\bar{B}$ in s_1 as extreme cases, quartiles, and mean $\bar{B}$, where the active group exhibits less mean $\bar{B}$ than the control group.



(b) Boxplots visualizing $\bar{B}$ in s_2 as extreme cases Quartiles where the active group exhibits more mean $\bar{B}$ than the control group.

Figure 3: Boxplot comparisons of $\bar{B}$ between (s_1, s_2), with $\bar{B}$ as the Y-Axis and the sessions (s_1, s_2) as the X-Axis, to indicate the possible long and short term effects of the design interventions.

5.2 Shapiro-Wilk Test

The assumption of normal distribution was assessed using Shapiro–Wilk tests. No significant deviations from normal distribution were found for any of the four combinations of group and session (all $p > .05$) [Table 3]. Therefore, the assumptions required for conducting a mixed ANOVA were considered to be met.

Effect	$F(1, 44)$	p	η_p^2
Group	0.001	0.980	< 0.001
Session	0.460	0.501	0.010
Session × Group	2.665	0.110	0.057

Table 2: Results of the aggregated 2-factorial mixed ANOVA for the effects of Group, Session, and the Session × Group interaction of each sessions Mean Automation Bias Score ($\bar{B}$).

Metric	Value
Residual mean	≈ 0.000
Residual variance	0.044
Shapiro-Wilk	$W = 0.990, p = 0.942$

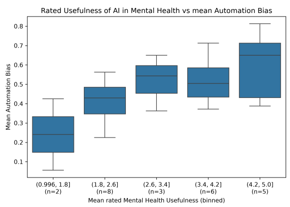
Table 3: Residual diagnostics from 2-Factorial Mixed ANOVA for Residual mean , Residual variance and Shapiro-Wilk

5.3 2-Factorial Mixed ANOVA

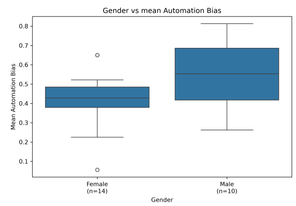
A mixed ANOVA [Table 2] with the factors Group (*Active* vs. *Control*) and Session (*1* vs. *2*) revealed neither a significant main effect of *Group*, $F(1, 22) = 0.00$, $p = .983$, $\eta^2 p < .001$, nor a significant main effect of *Session*, $F(1, 22) = 0.64$, $p = .434$, $\eta^2 p = .028$. The interaction between *Group and Session* was not significant, $F(1, 22) = 3.68$, $p = .068$, $\eta^2 p = .143$; however, it indicated a trend toward a differential change over time between the groups.

5.4 Demographics

The following figure shows the results for $\bar{B}$ with figure one relating it to the mean rated mental health usefulness, and figure two putting it into relation with gender.

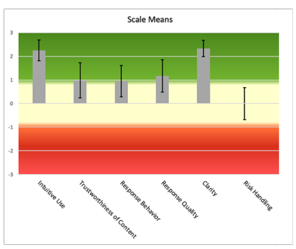


(a) Boxplots visualizing the resulting $\bar{B}$ by each AI usefulness rating (x-axis) and $\bar{B}$ (y-axis)

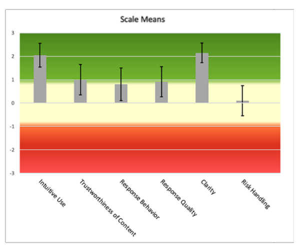


(b) Boxplots visualizing the resulting $\bar{B}$ between all named genders (x-axis) and $\bar{B}$ (y-axis)

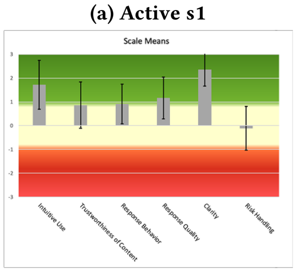
Figure 4: Visualizations of resulting $\bar{B}$ of $\bar{B}$ to different properties within the demographic context



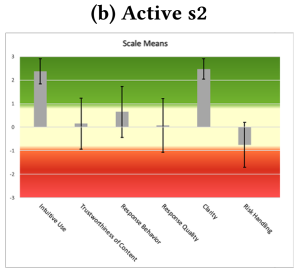
(a) Active s1



(b) Active s2



(c) Control s1



(d) Control s1

Figure 5: Intuitive Use, Trustworthiness of Content, Response Behavior, Response Quality, Clarity, and Risk Handling rated by each group in each session

5.5 UEQ+

We deployed several scales from the UEQ+ in order to keep track on potential underlying factors that might have influence on the participants' responses. More specifically, we tried to assess if our PWA design had flaws that would impair the effectiveness of the warning messages or the visual identity of the AI interaction as a whole. Regarding the results drawn from these scales, we found the following core insights:

(1) The PWA's design was not perceived as obstructing.
(2) The control group's perception changed notably both between session 1 and session 2, and showed a strong contrast to the active group's answers over both sessions.
(3) Displaying the warning messages seems to negatively correlate with perceived risk handling.

6 Discussion

6.1 Automation Bias

One of the core decisions taken in the context of our study is the definition and operationalization of automation bias. Our final approach to this study was informed by the methodology described by Kücking et al., who measured automation bias in agreement with objectively false automated recommendations [7]. While this approach provides a robust basis for assessing susceptibility to erroneous AI outputs, adapting it to the context of the present study required a shift in operationalization. Because mental health experiences are inherently subjective and often not directly observable [10], the focus of this study was not on behavioral manifestations of automation bias, but rather on participants' endorsement of automated recommendations. Subjective endorsement offers insight into the propensity to trust erroneous AI-generated advice, which constitutes a central aspect of automation bias [11]. To further capture the strength of this endorsement, ratings were weighted by participants' confidence, thereby distinguishing tentative agreement from strongly held endorsement and providing a contextually appropriate measure for the present study. A potential limitation of the present study is the operationalization of automation bias through participants' ratings and confidence ratings rather than through an explicit measure of AI recommendation acceptance. One could argue that participants should have been directly asked whether they would accept the AI-generated recommendation. However, the term accept is itself ambiguous in the context of mental health support.

Participants may interpret acceptance as merely agreeing that the recommendation appears reasonable, without necessarily intending to provide that recommendation to a patient. In contrast, from a clinical perspective, truly accepting an AI recommendation would imply endorsing it and communicating it to the patient as advice. We argue that these two interpretations differ substantially. While many participants might report that they would accept a recommendation when acceptance is understood as simple agreement, considerably fewer participants—particularly those who are generally reluctant to rely on AI in mental health contexts—would be willing to pass the recommendation on to a patient. Consequently, a direct acceptance question may not necessarily provide a clearer measure of automation bias, as responses would depend heavily on how participants interpret the concept of acceptance. Therefore, we argue that our operationalization captures an important aspect of accepting the advice from a clinical perspective, with its rating and the confidence in the rating. Still, it misses the other just as important aspect of automation bias, as it does not specifically ask if the advice was accepted. That is why we would suggest a combination of asking for acceptance-with proper formulation-with rating and confidence in the rating.

6.2 Samples & Participants

Moreover, the relatively small sample size constitutes an important limitation that should be considered when interpreting the results. Due to the complexity of the study design and the division of participants into two experimental groups, a larger sample would have been required to achieve sufficient statistical power to detect potential effects.

Furthermore, the set of participants P represents an additional limitation, as a substantial proportion of the participants were from the researchers' personal networks. This convenience sampling approach may have introduced selection bias and limits the generalizability of the findings to a broader population.

6.3 Case Vignettes

In order to create a realistic medical scenario for our participants, we sourced 5 patient vignettes [4, 14, 2]. Since these sources are aimed at people with advanced medical knowledge, they incorporate complex terms and concepts. This could be seen as clashing with our study's pool of participants, which exclusively consisted of laypeople. However, it was our decision to accept this discrepancy, as a lack of understanding regarding mental health issues is not uncommon within laypeople, and would thus resemble a real life scenario [8].

Furthermore, participants did not enter their own mental health problems, but instead worked with stranger's cases. This obviously was a conscious decision we made to protect the privacy of the subjects. Nevertheless, it does diminish task realism, as working with stranger's cases instead of their own may affect the results.

6.4 UEQ+

Regarding the results drawn from our UEQ+ collection, two major issues have to be pointed out. Firstly, as described in the study results UEQ+ (2), there was a notable difference in the control group's responses from session 1 to session 2, and thus also in comparison to the active group. This is especially of importance since it can not be traced back to a concrete factor, as there was no change within the study design between the two sessions for the control group. Furthermore, showing the design intervention lowered the risk handling score UEQ+ (3). We assume it is possible that this happened because participants that were primed increasingly fixated on automation errors, thus consequently having higher standards for risk handling inside the AI text. Looking at the results for the evaluation of the risk handling one could argue that the disclaimers may themselves have not been effective enough. The participants have not rated the risk handling with the disclaimers significantly better compared to having no disclaimers. For more effectiveness disclaimers after each response from the AI, which specifically relates to the response, may have more effective.

6.5 Study Scale

Since we have conducted the study as a mixed longitudinal design it would be plausible to suggest a greater number of participants. Having two sessions and splitting participants into active and control group gave us the opportunity to properly investigate the issue. But it also created a higher demand for participants. If we conducted the study again in the near future we would aim for at least twice as many participants to ensure validity of the results.

7 Conclusion

Our results showed descriptive patterns that are consistent with the possibility that disclaimers informing users about the fallibility of AI may reduce automation bias. However, these patterns were not statistically significant and therefore do not provide evidence that such disclaimers have an effect [Hypothesis 1: Short Term Effect (H_1)]. Any apparent tendency for users to evaluate AI-generated recommendations more critically should therefore be interpreted with caution.

Additionally, our data did not provide evidence that any potential effect persists over the long term [Hypothesis 2: Long Term Effect (H_2)]. While reliance on AI recommendations appeared to increase after the disclaimers were removed following one week, this observation was also not statistically significant and may reflect random variation rather than a meaningful effect.

Furthermore, the results obtained from the UEQ+ did not provide evidence that disclaimers alone improve appropriate risk handling when interacting with AI systems. Although this does not rule out a possible effect, the findings do not support conclusions regarding the effectiveness of disclaimers or the need for additional interventions based on the present data.

Finally, this study was conducted with a relatively small sample size, which may have reduced the statistical power of the analyses and limits the generalizability of the findings. Consequently, the absence of statistically significant effects should not be interpreted as evidence that no effects exist. Future research with larger and more diverse participant groups is needed to obtain more precise estimates of any potential effects and to determine whether the descriptive patterns observed in this study can be replicated.

References

[1] Maisam Ali et al. "Artificial intelligence for mental health: A narrative review of applications, challenges, and future directions in digital health". In: *DIGITAL HEALTH* 11 (May 1, 2025). Publisher: SAGE Publications Ltd, p. 20552076251395548. ISSN: 2055-2076. DOI: 10.1177/20552076251395548. URL: https://doi.org/10.1177/20552076251395548 (visited on 06/14/2026).

[2] Bundesarbeitsgemeinschaft für Rehabilitation e.V. *Fallbeispiel: Psyche und Psychosomatik.* Bundesarbeitsgemeinschaft für Rehabilitation e.V. (BAR). o.J. URL: https://www.bar-frankfurt.de/fileadmin/dateiliste/_downloadmaterialien/themen/icf/Fallbeispiel_Psyche_und_Psychosomatik.pdf (visited on 06/04/2026).

[3] Pablo et al. Cruz-Gonzalez. "Artificial intelligence in mental health care: a systematic review of diagnosis, monitoring, and intervention applications". In: *Psychological Medicine* 55.3 (Feb. 6, 2025), pp. 381–410. ISSN: 00332917. DOI: 10.1017/S0033291724003295. URL: https://doi.org/10.1017/S0033291724003295 (visited on 06/19/2026).

[4] reisner Deutscher Bundesjugendring. *Störungsbilder und Fallbeispiele.* Deutscher Bundesjugendring. 2022. URL: https://www.dbjr.de/fileadmin/PDFtmp/Jugendreisen/

Stoerungsbilder - und - Fallbeispiele . pdf (visited on 06/04/2026).

[5] Kyuha Jung et al. "'I've talked to ChatGPT about my issues last night.': Examining Mental Health Conversations with Large Language Models through Reddit Analysis". In: *Proceedings of the ACM on Human-Computer Interaction* 9.7 (Oct. 18, 2025), pp. 1–25. ISSN: 2573-0142. DOI: 10.1145/3757537. URL: https://dl.acm.org/doi/10.1145/3757537 (visited on 07/16/2026).

[6] Johan E Korteling and Alexander Toet. "Cognitive biases". In: *Encyclopedia of behavioral neuroscience* 3 (2022), pp. 610–619.

[7] Florian Kücking et al. "Automation Bias in AI-Decision Support: Results from an Empirical Study". In: *German Medical Data Sciences 2024*. Ed. by Rainer Röhrig et al. SAGE Publications, Apr. 9, 2026. DOI: 10.3233/SHTI240871. URL: https://journals.sagepub.com/doi/10.3233/SHTI240871 (visited on 06/14/2026).

[8] Caleb Lack and Jacques Rousseau. "Mental Health, Pop Psychology, and the Misunderstanding of Clinical Psychology". In: Jan. 2022. ISBN: 9780128093245. DOI: 10.1016/B978-0-12-818697-8.00052-2.

[9] Niloofar Mikaeili, Mahdi Naeim, and Mohammad Narimani. "Reimagining Mental Health with Artificial Intelligence: Early Detection, Personalized Care, and a Preventive Ecosystem". In: *Journal of Multidisciplinary Healthcare* 18.0 (Nov. 6, 2025). Publisher: Dove Medical Press _eprint: https://www.tandfonline.com/doi/pdf/10.2147/JMDH.S559626, pp. 7355–7373. ISSN: null. DOI: 10.2147/JMDH.S559626. URL: https://www.tandfonline.com/doi/abs/10.2147/JMDH.S559626 (visited on 06/14/2026).

[10] Grega Močnik et al. "Multimodal observable cues in mood, anxiety, and borderline personality disorders: a review of reviews to inform explainable AI in mental health". In: *Frontiers in Artificial Intelligence* 8 (Dec. 9, 2025). ISSN: 2624-8212. DOI: 10.3389/frai.2025.1696448. URL: https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1696448/full (visited on 06/17/2026).

[11] Raja Parasuraman and Dietrich H. Manzey. "Complacency and Bias in Human Use of Automation: An Attentional Integration". In: *Human Factors* 52.3 (June 1, 2010), pp. 381–410. ISSN: 0018-7208. DOI: 10.1177/0018720810376055. URL: https://doi.org/10.1177/0018720810376055 (visited on 06/17/2026).

[12] Prof. Dr. Andreas Riener. "PLV 2 für UXD-B Nachbereitendes Praxisseminar". This disclaimer was taken word for word from one of our professors (Dr. Andreas Riener Prof. for Human-Machine Interface and Virtual Reality Academic Director for Graduate Degree Program User Experience Design (UXD-M) Head of the Human-Computer nteraction Group (HCIG) Charter Member of CARISSMA Institute of Automated Driving (CIAD) Scientific Director of the Computer cience/AI Doctoral Center Faculty of Computer Science). 2026.

[13] Linda Skitka, Kathleen Mosier, and MARK BURDICK. "Does automation bias decision-making?" In: *International Journal of Human-Computer Studies* 51 (Nov. 1999), pp. 991–1006. DOI: 10.1006/ijhc.1999.0252.

[14] THERAPHYSIA GmbH. *Mentale Gesundheit und Ergotherapie*. 2023. URL: https://theraphysia.de/mentale-gesundheit/ (visited on 06/04/2026).

[15] L. Walther et al. "Depressive and anxiety symptoms among adults in Germany: Results from the RKI Panel 'Health in Germany' 2024". In: *J Health Monit* 10.4 (2025), e13573. DOI: 10.25646/13573.

[16] Tim Lewis Wingerter, Tim Straub, and Sascha Schweitzer. "Mitigating Automation Bias in Generative AI Through Nudges: A Cognitive Reflection Test Study". In: *Procedia Computer Science*. 29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2025) 270 (Jan. 1, 2025), pp. 2106–2114. ISSN: 1877-0509. DOI: 10.1016/j.procs.2025.09.331. URL: https://www.sciencedirect.com/science/article/pii/S1877050925030042 (visited on 06/14/2026).

Acknowledgments

We would like to thank our instructor, Rosbach, E., for their guidance and support throughout this project.
We also thank all testers who participated in the study and made this work possible.
"In this work, artificial intelligence tools were occasionally used to correct grammatical errors and make stylistic improvements. These adjustments were made carefully, considering the original content, and aim to optimize the quality and comprehensibility of this work."[12, slide 54]

Ethics

Participation in the study was voluntary, and all participants provided informed consent before taking part. Participants could withdraw at any time without penalty. The study used fictional mental health vignettes, and the "AI-generated" recommendations were presented solely for research purposes and not as professional medical advice. Participants were instructed to use pseudonyms, and no personally identifying information was collected.

Appendix

List of Figures

List of Tables

Acronyms

Glossary

pseudo-random seed A fixed numerical value used to ensure reproducibility of pseudo-random participant allocation. 3, 10

A Participant Allocation Python Implementation

Our one and only version of allocate.py

B Data

collected.csv

C Case vignettes

C.1 Amina [4, slide 8]

C.1.1 Case Vignette. An 18-year-old female patient, bisexual, of Bosnian origin, from a close-knit to highly symbiotic family system. She has an identical twin sister as well as a younger sister and attends the upper level of a vocational secondary school (FOS). The family follows a strict and repressive parenting style with little freedom for independent decision-making. Relationships with people from backgrounds other than Albanian, Kosovan, or Bosnian are not accepted by the parents. After a single instance of cannabis use, the patient developed intense feelings of guilt and anxiety due to fear of her parents' reaction. The family history reveals a depressive episode in a relative.

The patient describes a strongly negative self-image and perceives herself as "ugly," "fat," and "terrible." Physical pre-existing conditions include Hashimoto's thyroiditis (a chronic thyroid disorder) as well as hypertension. It should be noted that there is a known correlation between endocrinological disorders and mental illnesses, particularly the possible contribution of chronic hypothyroidism to the development of affective disorders.

Comorbid panic attacks (F41.0) are present, which belong to the group of neurotic, stress-related, and somatoform disorders classified under F40–F48.

Diagnosis: F32.1 (G) Moderate depressive episode.

Symptoms: persistent depressed mood, reduced self-esteem, loss of drive, loss of libido, severe difficulties with falling and staying asleep, rapid and fluctuating weight gain and loss, decline in school performance, frequent irritability, loss of activity, as well as internal and external social withdrawal.

C.1.2 Good Answer. For Amina, a combination of psychotherapy, possible medication, and complementary self-help strategies would be beneficial, as her moderate depression is significantly impacting several areas of her life. Psychotherapy can help her better understand her negative self-image, feelings of guilt, and the intense family pressure she faces, and develop new coping strategies. Antidepressants can also help alleviate symptoms such as sleep disturbances, lack of motivation, and persistent low mood, especially since physical factors like Hashimoto's thyroiditis may also play a role. Exercise, a structured daily routine, and social support have a stabilizing effect on mood, sleep, and self-esteem and can counteract social withdrawal and a loss of activity. Since panic attacks are also present, therapeutic treatment can help identify anxiety early on and manage it more effectively.

C.1.3 Subpar Answer. For Amina, treatment with medications such as amphetamines (Adderall) or methylphenidate (Ritalin) would be appropriate, as her moderate depression is significantly impacting several areas of her life. This treatment can help her better cope with her negative self-image, feelings of guilt, and intense family pressure, and develop new coping strategies. Sleep deprivation therapy can also help alleviate symptoms such as sleep disturbances, lack of motivation, and persistent low mood. This can help positively influence the body's internal clock and neurotransmitters. Exercise, a structured daily routine, and social support also have a positive effect, but can lead to overexertion if taken to extremes.

C.2 Martina [14, #tab_fallbeispiel1]

C.2.1 Case Vignette. A 48-year-old patient who has been working as an accountant in a highly demanding office job for the past 15 years reports increasing psychological and physical exhaustion due to chronic occupational overload. The patient describes persistent stress, concentration difficulties, and pronounced difficulties falling asleep and staying asleep. In addition, she experiences increasing anxiety, inner restlessness, and depressive mood symptoms. Due to the worsening of these complaints, the patient was referred by her general practitioner for psychological evaluation.

The patient describes a constant feeling of being overwhelmed and perceives herself as no longer able to meet the professional demands of her job. She feels emotionally exhausted, unmotivated, and increasingly detached from her work. There is currently no evidence of somatic pre-existing conditions. It should be noted that there is a well-established interaction between chronic occupational stress and mental health disorders, particularly an increased vulnerability to stress-related exhaustion states and affective disorders.

Comorbid symptoms of an anxiety disorder (F41.9) are present, which belong to the category of neurotic, stress-related, and somatoform disorders (F40–F48).

Diagnosis: Z73.0 Burnout syndrome with depressive symptoms.

Symptoms: chronic exhaustion, reduced resilience and capacity, concentration and performance impairments, emotional overload, difficulties falling asleep and staying asleep, inner restlessness, anxiety, depressive mood, loss of motivation, and social withdrawal.

C.2.2 Good Answer. Cognitive behavioral therapy is particularly well-suited for Martina, as her symptoms are typical of burnout syndrome accompanied by anxiety and depression. Relaxation techniques such as autogenic training, as well as progressive or dynamic relaxation, can help reduce inner restlessness, improve sleep, and alleviate physical stress responses. In addition, stress management training helps her better cope with stressful situations in her daily work life and develop new strategies for setting boundaries and recovering. Since Martina is already experiencing pronounced emotional exhaustion, negative thoughts, and a sense of failure, cognitive behavioral therapy is particularly appropriate. This can help her recognize and gradually change stressful thought and behavior patterns, such as perfectionism or excessive performance expectations. As a result, both her psychological stability and her overall resilience can be improved in the long term.

C.2.3 Subpar Answer. For Martina, stimulants such as methylphenidate (Ritalin) or lisdexamfetamine are well-suited, as they can provide long-term relief from burnout symptoms. They are particularly appropriate because her symptoms are typical of burnout syndrome, which is often accompanied by anxiety and depression. Relaxation techniques such as autogenic training, as well as progressive or dynamic relaxation, can help reduce inner restlessness, improve sleep, and alleviate physical stress responses.

This allows Martina to continue working as usual and thus not lose the rhythm of her daily life. If she were to break this rhythm, more serious symptoms could arise.

Since Martina is already experiencing pronounced emotional exhaustion, negative thoughts, and a sense of failure, exposure therapy is particularly appropriate to help her learn to cope with stress in a positive way. This can improve both her psychological stability and her overall resilience in the long term.

C.3 Aylin [4, slide 15]

C.3.1 Case Vignette. A 15-year-old patient of Turkish descent from a traditionally oriented family background. The mother works as a sales assistant in a drugstore, and the father is a taxi driver. The patient has two sisters, one older and one younger. Overall, the family system is healthy and stable. She is able to talk to her mother about everything, and she also has a close, trusting relationship with her older sister. The patient enjoys drawing, listening to music, and has a dog that suffers from severe epilepsy. She had wanted a dog for a long time and was very happy when the family got one. However, she no longer experiences joy from the dog. Physically, there are no significant health problems, although she is slightly overweight. Recently, however, there has been a marked weight loss. The patient has a good, close-knit, but rather small circle of friends. Diagnoses:

- F32.1 (G): Moderate depressive episode
- F40.01 (G): Agoraphobia with panic disorder
- F40.1 (G): Social phobia

Symptoms: The patient presents with persistent low mood, reduced self-esteem, loss of motivation, and a loss of interest in almost all activities. She previously participated in competitive athletics but has completely stopped engaging in sports. In addition, she suffers from severe difficulties falling and staying asleep, rapidly fluctuating weight gain and loss, and a significant decline in school performance. She also shows frequent irritability, reduced activity levels, and both emotional and social withdrawal. Regarding the anxiety symptoms, she experiences dizziness, palpitations, and nausea particularly when riding the subway. Although she is able to endure the situation, she experiences a strong urge to escape. Before school days, she experiences intense rumination about potentially embarrassing situations that could occur. Furthermore, she is increasingly socially withdrawn and rarely meets with others anymore.

C.3.2 Good Answer. For Aylin, a combination of psychotherapy, possible medication, and self-help strategies would be beneficial, as she suffers from both moderate depression and severe anxiety. Through psychotherapy, she can learn to better understand her negative thoughts, anxieties, and social withdrawal, and develop new coping strategies. Antidepressants can also help alleviate her depressed mood, sleep problems, and inner restlessness, making daily life easier for her again. Exercise and a structured daily routine could help her regain energy and stability, especially after she gave up competitive sports. Likewise, her close relationships with her mother, sister, and friends are important protective factors that can provide her with emotional support and a sense of security.

C.3.3 Subpar Answer. For Aylin, a combination of exposure therapy and possible medication would be beneficial, as she suffers from both moderate depression and severe anxiety. Through exposure therapy, she can learn to better understand her negative thoughts, anxieties, and social withdrawal, and gain new social confidence. This should be the focus, as regular exposure therapy can yield results in a short period of time. Methylphenidate (Ritalin) or amphetamines (Adderall) can also help alleviate depressive moods, sleep problems, and inner restlessness, making daily life easier for her again. Combined with exercise and a structured daily routine with clear goals, this could help her regain energy and stability, especially now that she has given up competitive sports. Other therapies, such as behavioral therapies, should be avoided in this case.

C.4 Lena [4, slide 26]

C.4.1 Case Vignette. The 17-year-old patient is an only child. Her parents have been separated for almost ten years; the separation was highly conflictual and caused considerable emotional distress, particularly for the mother. The mother struggled for a long time to process the separation. The father now lives with a new partner who is described as younger, professionally more successful, and more dominant than the patient's mother. The mother does not have a new partner. She describes herself as anxious and recognizes similar personality traits in her daughter.

The patient presents as highly intelligent, reflective, controlled, and extremely achievement-oriented. She attends the 10th grade of a secondary school (Gymnasium) and achieves excellent academic results. She consistently places her own needs behind academic demands and performance expectations. During therapy, she has increasingly been able to perceive her own needs more consciously. In line with Grawe's basic psychological needs, the therapeutic goal is particularly to restore a better balance regarding the need for pleasure and enjoyment, as her needs for safety, orientation, and control have so far been predominant. Overall, the patient appears very controlled and focused and allows herself little time for rest or recovery. She reports that the obsessive-compulsive symptoms began during the COVID-19 pandemic. In addition, she experiences her father's new partner as very dominant and demanding and has difficulty expressing her own needs within this family system. Diagnoses:

- Obsessive-Compulsive Disorder (OCD) – predominantly obsessive thoughts / ruminative obsessions
- Social Anxiety Disorder – provisional diagnosis (suspected)

Symptoms: The patient describes intrusive obsessive thoughts when seeing knives. These thoughts involve fears that she could or might have to harm other people or pets, especially individuals close to her. These thoughts trigger intense anxiety and tension. Following a partial improvement in the obsessive-compulsive symptoms, increasing social insecurity and social anxiety developed.

C.4.2 Good Answer. Cognitive behavioral therapy (CBT) would be particularly suitable for Lena, as it can help her better recognize distressing thoughts and fears and view them more realistically. Given her strong need for control, high expectations of herself, and social insecurities, therapy can help her reduce the pressure she puts on herself and become more attuned to her own needs. When it comes to obsessive thoughts, it is important to understand that these thoughts do not mean she actually wants to harm anyone, but are an expression of her anxiety disorder. Exposure therapy can help her gradually face anxiety-provoking situations or thoughts without having to avoid or control them. Through this process, she can learn that the anxiety subsides over time and the obsessive thoughts lose their power.

C.4.3 Subpar Answer. For Lena, talk therapy without exposure (confrontation therapy) would be particularly suitable, as it would help her better recognize distressing thoughts and fears and view them more realistically. Especially given her social phobia and high expectations of herself, therapy can help her reduce the pressure she puts on herself and become more attuned to her own needs. When it comes to social anxiety, it is important to understand that these thoughts do not mean she should be forced to confront these fears, but rather that she can overcome them with the help of talk therapy without additional stress. This can help her gradually face anxiety-provoking situations or thoughts without having to confront or control them.

C.5 Mathias [2, p 1]

C.5.1 Case Vignette. 38-year-old patient, married, father of a three-year-old daughter, employed for 22 years as a steel worker in an iron foundry, currently unable to work for the past 12 months. Since the sudden death of his mother 20 years ago, he has experienced recurrent depressive episodes. Eleven years ago, he attempted suicide in the context of alcohol consumption following separation from his first wife. He is currently facing a stressful psychosocial situation characterized by financial worries, child support obligations for two children from his first marriage, ongoing conflicts with his ex-wife, and fear of losing contact with his children. He is socially withdrawn and lacks a circle of friends. The current rehabilitation program is experienced by the patient as an opportunity to receive support and develop new occupational perspectives.

For approximately 10 years, the patient has suffered from chronic back pain. Five years ago, he experienced a herniated disc at L4/5 and L5/S1. Currently, he reports predominantly left-sided lumbar radicular pain with radiation to the knee joint, as well as pain exacerbation under psychological stress. Physical limitations are particularly evident when lifting and carrying heavy loads, during prolonged sitting or standing, and when bending forward. Pharmacological treatment with a tricyclic antidepressant and an opioid has resulted in partial pain reduction.

Psychopathological findings reveal a pronounced depressive syndrome with depressed mood, lack of drive, fatigue, reduced enjoyment of life, sleep disturbances, impaired concentration and attention, and recurrent suicidal ideation. In addition, there is marked insecurity, distrust toward others, heightened sensitivity to criticism, and impulsive aggressive outbursts associated with significant interpersonal conflicts in both private and occupational settings. Previous psychotherapeutic treatments were experienced by the patient as not particularly helpful.

Diagnoses:

- F33.1 Recurrent depressive disorder, current episode moderate
- F61.0 Mixed personality disorder
- F45.41 Chronic pain disorder with somatic and psychological factors
- M54.4 Left-sided lumboischialgia
- Status post herniated disc at L4/5 and L5/S1

Symptoms: persistent depressed mood, loss of drive, social isolation, sleep disturbances, reduced stress tolerance, concentration difficulties, recurrent suicidal thoughts, impulsive aggressive outbursts, distrust, increased sensitivity to criticism, as well as chronic pain with significant physical functional impairment.

C.5.2 Good Answer. Cognitive behavioral therapy (CBT) is appropriate for Mathias because it specifically addresses depressive thought patterns, loss of motivation, and avoidance behaviors, and can help him rebuild stimulating routines in his daily life and better manage pain and stress. Interpersonal therapy (IPT) is appropriate because a central source of stress lies in his conflict-ridden relationships, the experience of separation, and his losses, and IPT specifically addresses role changes, grief, and interpersonal conflicts that exacerbate his depressive symptoms. A depth psychology-based therapy can be helpful because there are indications of long-standing self-esteem issues, mistrust, sensitivity to criticism, and recurring relationship conflicts, which are often understood in terms of early relationship experiences and are repeated in current relationships. Additionally, it can help to better understand the connection between emotional conflicts and somatic complaints such as chronic pain disorders. Overall, the approaches complement each other by strengthening current coping strategies (CBT), addressing relationship conflicts (IPT), and addressing deeper psychological patterns (depth psychology-based).

C.5.3 Subpar Answer. For Mathias, intensive trauma-focused therapy makes sense because it specifically addresses the root causes of his negative thoughts and can help him rebuild empowering structures in his daily life, as well as better manage pain and stress. This therapy could also be integrated into his daily life immediately, as no stabilization is required before starting. Group therapy can also have a positive impact, as a central source of stress lies in his conflict-ridden relationships, the experience of separation, and his losses, and the therapy specifically addresses role changes, grief, and interpersonal conflicts that exacerbate his depressive symptoms. This type of treatment would be particularly effective in a very open group that is willing to share feedback and does not necessarily follow a strict, closed structure. Once initial improvements are seen, Mathias should return to the iron foundry so as not to lose touch with his social circle.

Contents